HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79SU

Survival Models

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2023

Duration: 2 hours

Attempt all questions

Total Marks: 60

Attached: No additional documents attached

Supplied: Formulae and Tables for Examinations of the Institute and Faculty of Actuaries (2002)

Calculator Permitted: Yes

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. The data below show the time (in weeks) until a person dies after testing positive for a terminal illness. The + symbol indicates that the observation was censored.

1 3+ 5+ 6 7+ 11 11+ 13 13+ 13+ 13+ 14+ 14+ 17 19+

(a) Calculate the Kaplan-Meier estimate of the survivor function $\hat{S}(t)$ for these 15 individuals.

[4 marks]

(b) Briefly explain an assumption we must have for the censoring mechanism and give a reason why it may not hold.

[2 marks]

(c) Explain why $\hat{S}(20)$ is undefined for this dataset.

[1 marks]

[Total 7 marks]

2. (a) Explain how a cross-sectional study can be used to build a model of mortality over a range of ages.

[4 marks]

(b) Give two important features to be aware of when conducting a cross-section study.

[2 marks]

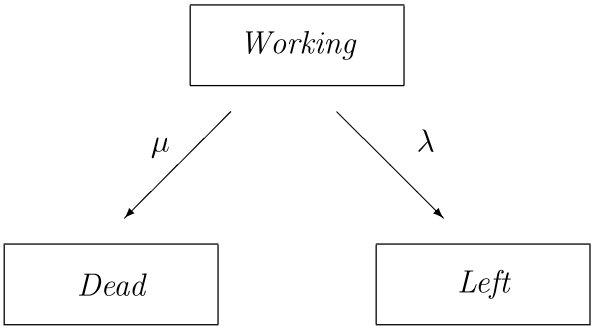
(c) Describe how a series of cross-sectional studies can be used to forecast future mortality.

[2 marks]

[Total 8 marks]

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3. Consider the following 3-state Markov model for working for a company, leaving the company and dying with constant transition intensities μ and λ.



(a) Explain in general what assumptions are needed to apply a Markov model with constant transition intensities.

[4 marks]

(b) Write down the full log-likelihood for μ and λ for the above 3-state model, explaining any notation that you have used.

[3 marks]

(c) Use your answer for (b) to derive an estimate $\hat{\lambda}$ for λ.

[2 marks]

(d) You are given data for a company where you decide to apply this model to their 40 employees, assumed to be independent and with the same transition rates. This company is unsuccessful causing all living employees to lose their jobs. It is found using this data that $\hat{\mu} = 0.01$ per annum and there is a total observed waiting time of 6000 months across all employees. Provide an estimate for $\hat{\lambda}$.

[2 marks]

[Total 11 marks]

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4. In a mortality investigation, the following data have been recorded in respect of lives observed between the ages of x and $x+1$, where

- a_i is the time after exact age x when the ith life came under observation;
- b_i is the time after exact age x when the ith life will be censored;
- $d_i = 1$ if the ith life dies before age $x + b_i$;
- $d_i = 0$ if the ith life survives to age $x + b_i$.

Suppose $x = 70$ and let $q_{70} = q$. The data are as follows:

Life i	a_i	b_i	d_i
1	0	1	1
2	0.5	1	0
3	0	0.5	1
4	0.25	0.75	0
5	0.25	1	1

(a) Using the uniform distribution of deaths assumption and the binomial model of mortality, show that the log-likelihood for these data is given by

$$\ell(q) = 3\log(q) + \log(1-q) + \log(1-0.75q) - \log(1-0.5q) - 2\log(1-0.25q).$$

[4 marks]

(b) Verify that the maximum likelihood estimator for q for this likelihood is $\hat{q} = 0.7368$.

[2 marks]

(c) Calculate an approximate standard error for the estimate found in (b).

[3 marks]

(d) Now find an estimate for the maximum likelihood estimate for μ assuming the Poisson model of mortality, assuming that all deaths which occur do so at the midpoint of each observation period.

[2 marks]

(e) Compare your answers for (b) and (d).

[2 marks]

[Total 13 marks]

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5. (a) A large mortality study has been performed and the data has been graduated using the formula

$$\mu_x = \frac{A + Bc^x}{Ec^{-x} + 1 + Dc^x}.$$

(i) It is found that $c = 1.12$. Describe a possible effect of the parameters D and E.

[2 marks]

(ii) Based on your answer to (i), what property of a dataset could cause the estimates of D and E to be unreliable?

[1 marks]

(b) Describe circumstances where graduating against a standard table may be appropriate.

[3 marks]

(c) Explain briefly how customers holding multiple policies can affect estimates in mortality modelling.

[2 marks]

[Total 8 Marks]

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6. (a) Briefly describe the differences between life year rate intervals and calendar year rate intervals.

[4 marks]

(b) State the Principle of Correspondence used when estimating mortality rates.

[2 marks]

(c) You have the following numbers of deaths observed at age 45 next birthday during the calendar years 2016, 2017 and 2018.

2016	2017	2018
32	33	34

You also have the following census data at age x last birthday, observed on 1 January in each of 2016, 2017, 2018 and 2019.

Age	1/1/2016	1/1/2017	1/1/2018	1/1/2019
44	7,645	8,430	8,224	7,331
45	6,890	6,889	8,822	8,345
46	7,277	7,763	7,342	8,111

(i) State, giving reasons, what definition of age you would choose for the estimation of crude forces of mortality $\hat{\mu}_x$

[2 marks]

(ii) Calculate an estimate for $\hat{\mu}_{45}$ and explain the exact age at which this estimate would apply.

[3 marks]

(iii) Using your result for (ii), estimate q_{45} and state the exact age to which this estimator would apply.

[2 marks]

[Total 13 marks]

END OF PAPER